# HONGYU HÈ

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( *Version: September 2026* | *indicates a hyperlink* )

## EDUCATION

**Princeton University — *PhD in Electrical Engineering*** *Aug. 2024 – present*

- Committee: Maria Apostolaki (advisor), Sharad Malik, Aarti Gupta *United States*
- GPA: 3.95/4.0 (8 graduate-level courses in EE and CS)

**ETH Zürich — *MSc in Computer Science*** *Aug. 2021 – Jul. 2024*

- Direct Doctoral Scholarship (5 out of 230 CS master's students in 2022) *Switzerland*
- Thesis awarded highest grade: 6.0/6.0 (Advisor: Gustavo Alonso)
- Associated Researcher at ETH AI Center

**Vrije Universiteit Amsterdam — *BSc (Hons) in Computer Science*** *Aug. 2018 – Jul. 2021*

**Universiteit van Amsterdam — *BSc Minor in Mathematics*** *Sep. 2019 – Jan. 2021*

- GPA rank: 1st of 180; national top 8 in CS (KHMW 2020) *The Netherlands*

**Tianjin University — *BA in Law*** (unfinished) *Sep. 2016 – Aug. 2017*

- Top 0.2% among 153,276 regional liberal arts students in National College Entrance Exam, "Gaokao" *China*

## INDUSTRY EXPERIENCE*

**Microsoft — *Research Intern*** | Networking Research Group *Jun. 2026 – Sep. 2026*

**Curieo AI (sold) — *ML Research Fellow*** | remote, part-time *Jul. 2024 – Oct. 2024*

**IBM Research — *Engineering Intern*** | Geodata Team | part-time *Apr. 2024 – Jun. 2024*

**Apple Inc. — *AI/ML Intern*** | Visual Intelligence Team | *(return offer)* *May 2023 – Oct. 2023*

**Oracle Labs — *Graal Cloud Native Intern*** | Serverless Team | part-time | *(return-intern offer)* *Feb. 2023 – May 2023*

**Dexter Energy Services — *BSc Thesis Intern*** *Jun. 2021 – Sep. 2021*

**Huawei Technologies (Amsterdam Research Center) — *R&D Intern*** | Search Team *Mar. 2021 – Jun. 2021*

**Picnic Technologies — *Software Developer*** | Store Team | part-time *Sep. 2020 – Mar. 2021*

**— *Backend Engineer Intern*** | Mentor: Sander Mak | *(return offer)* *Jun. 2020 – Sep. 2020*

## TEACHING

**Princeton University** | 20 hrs/wk (head TA) | Computer Networks *2025*

**ETH Zürich** | 14 hrs/wk | Distributed Systems Lab, Big Data (my session recordings) *2023, 2024*

**Vrije Universiteit Amsterdam** | 8–16 hrs/wk | Graph Theory, Linear Algebra, C++, Calculus I *2019, 2020*

## TECHNICAL SKILLS

**Programming & Scripting**: C/C++, Python, (System)Verilog, Golang, Shell, Scala, Java, SQL, LaTeX
**Software & Tools:** Kubernetes, Xilinx Vivado/Vitis suite, JAX, PyTorch, Linux, Git, Docker, Knative, Apache Spark, SLURM, STM32 (ARM Cortex-M3), MongoDB, PostgreSQL, Hadoop, HBase, Spring5, Qt5

*Industry recommendations: hhy.ee.princeton.edu/rec/

## PUBLICATIONS

### Preprint

[14] H. Hè, M. Apostolaki. *Let AI Agents Translate Networks, Not Reason About Them*. [PDF]

### Peer-reviewed paper

[13] H. Hè, M. Jin, M. Apostolaki. *Worst-Case Discovery and Runtime Protection for RL-Based Network Controllers*. USENIX NSDI 2027 (to appear) [PDF]

[12] H. Hè*, A. Krentsel*, S. Ratnasamy, M. Apostolaki. *Invariant Discovery for Networked Systems*. ACM HotNets 2026 (to appear) [PDF]

[11] C. Constantine, H. Hè, M. Apostolaki. *No Hyperscaler? No Problem. Your Students also Break (and Fix) Networks*. ACM HotNets 2026 (to appear)

[10] M. Jin, H. Hè, M. Apostolaki. *Cross-Flow Correlations Survive Synthesis: Measuring Source-Level Privacy Leakage in Synthetic Network Traces*. NDSS 2027 (to appear) [PDF]

[9] H. Hè, M. Jin, M. Apostolaki. *Making Logic a First-Class Citizen in Generative ML for Networking*. USENIX NSDI 2026 [code] [PDF]

[8] H. Hè, M. Apostolaki. *Just-in-Time Logic Enforcement: A new paradigm of combining statistical and symbolic reasoning for network management*. ACM HotNets 2025 [code] [PDF]

[7] A. Zhou, C. Costic, H. Hè, A. Ghalayini, A. Kabbani, M. Apostolaki. *Mitigating Inter-datacenter Incast with a Proxy: The shortest path is not necessarily the fastest*. ACM HotNets 2025 [PDF]

[6] B. McKinzie, Z. Gan, J. Fauconnier, ..., H. Hè, ..., P. Grasch, A. Toshev, Y. Yang. *MM1: Methods, Analysis & Insights from Multimodal LLM Pre-training*. ECCV 2024 **(first flagship multimodal LLM from )** [PDF]

[5] H. Hè, M. Friedman, T. Rekatsinas. *EnergAt: Fine-Grained Energy Attribution for Multi-Tenancy*. ACM HotCarbon 2023 **(adopted by Trycarbonara)** [media] [code] [PDF]

[4] B. Kruit, H. He, J. Urbani. *Tab2Know: Building a Knowledge Base from Tables in Scientific Papers.* ISWC 2020. [code] [PDF]

### Patent

[3] S. Kakarla, R. Beckett, H. Hè. *Autoformulation of Executable Cost-Saving Rules from Cloud Billing Policies with AI*. Under review.

### Degree thesis

[2] H. Hè. *FPGA-based SmartNIC for Distributed Machine Learning*. 2024 **(best grade: 6.0/6.0)** [MSc thesis]

[1] H. He. *How Can Datacenters Join the Smart Grid to Address the Climate Crisis? Using simulation to explore power and cost effects of direct participation in the energy market*. 2021 **(Amsterdam Thesis Award)** [BSc thesis]

## OTHER ACHIEVEMENTS

- ♦ Nominee for Princeton Teaching Award (3 nominees/department annually, see letters from my students here) *2026*
- ♦ Princeton Natural & Artificial Mind Fellowship in AI *2025*
- ♦ Princeton University Administrative Fellowship in CS *2024*
- ♦ Nominee for ETH Medal (MSc) *2023*
- ♦ Danish Government scholarship [€ 28,260] *2021*
- ♦ Leiden Excellence scholarship [€ 15,000] *2021*
- ♦ GRE 334 (verbal: 161, quant: 170, writing: 3) *2021*
- ♦ GLOBE scholarship (ranked 1st out of 900+ ) [€ 1,250] *2020*
- ♦ ACM-ICPC: Amsterdam Algorithm Programming Preliminaries (AAPP) with my team, ranked 1st *2020*
- ♦ ACM-ICPC: Benelux Algorithm Programming Contest (BAPC) with my team, ranked 9th *2019*
- ♦ Young Talent Award by *The Royal Holland Society of Sciences (KHMW)* *2019*
- ♦ Ranked 1st in Operating Systems (Cristiano Giuffrida) and Networks (Andrew Tanenbaum) courses *2019*

## LANGUAGES

**English**: TOEFL (speaking): 28, IELTS: 9 | **German**: B1 | **Dutch**: Basic | **Mandarin**: PSC: Level 1-B, regional max.